



Fig. 1: Samsung 970 Pro and 970 Evo (Credit: <https://wccfttech.com>)

input or output activities. It does not prioritise based on individual application. This allows the SSD array to provide improved data access. To provide enhanced performance, the SSA employs the SSD to store data in block forms. An SSD array can be accessed through multiple physical hosts, so is easy to set up.

A rich software layer governs the SSD arrays and controls a range of activities like compression, replication and snapshots. The type of software determines the functionalities of SSAs. Software is controlled by storage controllers to control data storage. SSAs have a range of advantages such as easy to set up, low latency and storage efficiency. These are also easy to integrate with data centres and solutions.

NAND is a non-volatile flash memory used in SSDs. It can retain data even after the power is turned off. Single-level cells (SLCs), multi-level cells (MLCs) and triple-level cells (TLCs) are the three different types of NAND.

SLCs have faster write speeds, higher program and erase cycles, lower power consumption but are the most expensive. These offer multiple solutions for enterprise use.

MLCs have low write speeds, higher data density, low cost, and low program and erase cycles. These are used for consumer-grade applications.

TLCs have low write speeds, low program and erase cycles, and high storage density. These are also for consumer-

grade solutions, and are the cheapest.

Vikas Jain, team leader - electrical schematics and apparatus, Cyient, says, "The technology we are using for SSA devices is NAND flash (SLC, MLC or

TLC). Our primary focus is on endurance, safety of data from EMI/EMC interference, magnetic field, data security and good to work in harsh regions like locomotive/railway engines or metro trains. We use Ethernet, RS485, CAN, MVB, I2C and USB for firmware porting/data backup."

Gartner Magic Quadrant for SSAs

IT and development and operation companies make the best SSAs in the world, as ranked by Gartner Magic Quadrant visionaries. Tintri VMstore T5000 series was among the top-named SSAs by Gartner.

SSA Magic Quadrant is essentially an all-flash array. It has some unique requirements for an array to qualify into the solid-state chart. It is for SSAs that are on the edge of performance

that focus on revenue-generating applications. Most products listed on the solid-state chart are from vendors who claim that the listed arrays are acceptable replacements for hard-disk-based systems.

It is expected that within the next 12 months, SSAs will improve in performance by a factor of 10, and double in density and cost-effectiveness, therefore changing the dynamics of the storage market.

Strategic planning assumption says that by 2021, 50 per cent data centres will use SSAs for high-performance computing and Big Data workloads. These systems include a data management software and an OS optimised for solid-state technology.

Design flow of SSDs

Jain says, "The main control unit is interfaced with a flash memory IC having SSD blocks. The interface is normally serial type, which may be high-speed SPI or I2C. For retrieving data from memory, a USB interface is used, or an indirect Ethernet connection to log data to the Web server."

Solid-state storage (SSS) technology infrastructure is based on standards that interact with the host

OS. Standards define how components communicate and connect. This drives volume and enables interoperability.

As shown in Fig. 4, there are two types of SSS devices: one that integrates the host bus adaptor (HBA) and connects directly to the host bus, and the other with an HDD-type interface, which connects via a separate HBA card.

Driver interface is a key enabler of a peripheral device and a standard software driver for the host OS. Host Bus PCI Express is a widely-implemented host bus and an SSS device interface.

Open NAND flash interface has developed interface specifications



Fig. 2: Gartner Magic Quadrant for SSAs 2017 (Source: Gartner, July 2017)